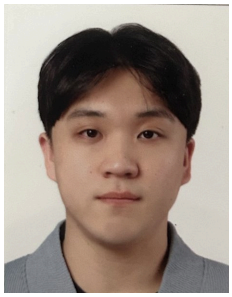
Hyunjun Ahn



AI / ML Engineer

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SUMMARY

AI engineer who takes models from raw data to systems that run every day. I designed, trained, and operate two production AI systems end to end: an F&B trend radar built on a distilled 4B LLM that outperforms a 30B general model at zero external API cost, and a reorder forecasting service that drives in-app restock alerts. I treat evaluation as the core of the work — frozen eval sets, time-ordered validation, and holdouts so that every "it got better" is a number, not a feeling. My foundation is C/C++ and Linux systems programming from the 42 curriculum, which shapes how I approach serving, observability, and data quality.

EXPERIENCE

Manwol · AI Engineer Sep 2026 – Present

B2B food-service e-commerce for cafés and bakeries · Suwon, South Korea

- Own the company's in-house AI systems from data collection and model training to serving, dashboards, and daily operation
- Shipped two production AI systems (Trend Radar, Reorder Forecasting — see Projects), both running unattended every day
- Built and launched Dessert Map, a Flutter map app for trending desserts, on the App Store and Google Play as a solo developer
- Built the in-app grocery mall and recipe features of the Manwol Check store-operations app, launched its owner community, and led UI/UX refactoring

Samseonghyeon History and Culture Museum · Researcher Mar 2025 – Dec 2025

Gyeongsan, South Korea

- Managed and analyzed artifact datasets; organized and maintained collection databases for historical records and exhibition documentation

PROJECTS

Trend Radar — Early Detection of Emerging F&B Trends · Manwol 2026 – Present

Python, Qwen3-4B, LoRA SFT, Knowledge Distillation, vLLM, scikit-learn, Prometheus, Grafana

- Built a daily pipeline that reads ~150K posts from 6 sources (Naver blog/café, Instagram, YouTube in ko/en/ja/zh, communities), extracts newly appearing food names with a local LLM, and tracks their search curves to flag trends before they go mainstream
- Distilled Qwen3-4B from teacher-model labels with rationales (Distilling Step-by-Step, LoRA SFT); the 7.5× smaller model beats a 30B general model on all 6 judgment attributes (food F1 0.905 → 0.947, café-fit F1 0.846 → 0.942) and lifts extraction F1 0.654 → 0.747
- Serves 4 vLLM containers on demand on an in-house GPU at zero external LLM API cost; captures 98% of names mentioned in 5+ posts and registered 27 of 28 real trends in the gold set
- Trend classifier on search-curve features reaches AUC 0.839 on a time-based holdout (growth-rate rule: 0.656); added source-span grounding, alias validation, and daily anomaly checks after tracing a 25× search-volume inflation to a bad alias

Reorder Forecasting — Predicting When Stores Buy Again · Manwol 2026 – Present

Python, XGBoost, TabPFN, Pandas, BigQuery, Cloud Run Jobs, Firestore, Cloud Functions, Grafana

- Predicts per store × item reorder date and quantity so the app can push restock alerts and pre-filled carts; runs daily as a Cloud Run Job → Firestore → Cloud Functions pipeline
- XGBoost cut reorder-date MAE by 34% (8.44 → 5.56 days) under 12-fold monthly walk-forward validation with all features recomputed as of prediction time to prevent leakage
- Re-framed the target from date error to late-alert rate after error analysis; a ratchet correction cut repeat misses 2.5% → 1.0% across 22,311 reorders, and a 5% no-alert holdout guards against self-confirming evaluation
- Built a cadence-type multi-agent router (TabPFN / quantile XGBoost) but kept the simpler model in production when the overall gain was under 0.4 pp — runtime 20s → 3.6s and PyTorch dropped from the image

Dessert Map — Store Finder App for Trending Desserts · Manwol 2026 – Present

Flutter, Python, Firebase Functions, Firestore, FastAPI, PostgreSQL, Naver / Kakao APIs

- Solo-built and shipped an iOS / Android app that maps where trending desserts are sold; a daily collector across 17 provinces creates stores only from verified place results and confirms sales with 2+ matching posts, removing false positives from name guessing
- Expanded to several dessert categories (up to 251 stores each) with crowd reports gated by 3+ distinct users and an admin review console

Manwol Check — In-app B2B Grocery Mall, Recipes & Community · Manwol 2026 – Present

Flutter, Firebase Cloud Functions, Cloud SQL (PostgreSQL), Firestore, Toss Payments, Remote Config

- Built in-app B2B ordering on top of an external store builder with no order-creation API: synced products / members via Cloud Functions, kept app orders in a PostgreSQL ledger feeding fulfillment, and implemented Toss payments, saved-card one-tap checkout, shipping-fee rules, self-cancel vs. return flows, and encrypted refund accounts
- Linked recipes to the mall — "buy ingredients" sheet and a "makes about N servings" estimate parsed from pack sizes (60g×10, 1kg ×3) with g↔ml conversion — and auto-filled smart carts at the reorder model's alert time
- Launched an owner community (channels, search, DMs, reporting / blocking, moderation dashboard) and refactored screens onto the SEED design system

Autobahn Truck Parking Availability Prediction · Applied Deep Learning 2026 – Present

Python, PyTorch, LightGBM, Kubernetes, TimescaleDB, Feast, Redis, Grafana, Flutter

- Forecasting rest-area parking availability from ~160K Toll Collect SID records (LightGBM and deep models vs. a seasonal baseline), served on Kubernetes with TimescaleDB, Feast, Redis, and Grafana MAE monitoring

Integrated Digital Cockpit · SEA:ME Capstone 2026

Xen Hypervisor, Yocto / BitBake, Raspberry Pi 4, Qt / QML, CAN Bus

- Built a mixed-criticality cockpit on a Raspberry Pi 4 with Xen Dom0 / DomU separation, integrating a digital cluster, Qt head unit, and ultrasonic parking distance control into a custom Yocto image, validated on an RC car

TECHNICAL SKILLS

AI / ML: LLM fine-tuning (LoRA SFT), knowledge distillation, vLLM, PyTorch, XGBoost, LightGBM, TabPFN, scikit-learn, Pandas, NumPy, time-series forecasting

MLOps & Data: Walk-forward validation, holdout / A/B design, Prometheus, Grafana, BigQuery, Cloud Run, Firestore, Feast, TimescaleDB, Redis, Docker, Kubernetes

Languages: Python, C, C++, TypeScript, Shell Script, OCaml

Systems: Linux, Yocto / BitBake, Xen Hypervisor, CAN Bus, Qt / QML, Argo CD, NGINX

EDUCATION

Korea National Open University – PRIME College · *B.S. in Artificial Intelligence* Mar 2026 – Present

42 Wolfsburg – SEA:ME Program · *Automotive Embedded AI · Sponsored by Volkswagen Korea* Jan 2026 – Present

42 Gyeongsan Common Core · *C, C++, Unix, algorithms, systems programming* Jan 2024 – Jan 2026

Yeungnam University · *B.A. in History* Mar 2018 – Feb 2024

AWARDS

Rising Star Award · 2025 Hackathon by 42 in Asia Nov 2025

CDG · 42 Gyeongsan

- Built an intelligent document processing pipeline for multilingual government documents: PaddleOCR-VL (OCR), XLM-RoBERTa (classification), Ollama (summarization), containerized with Docker

Grand Prize · ESG Living Lab Matching Hackathon Dec 2024

Ministry of Education, Republic of Korea

- Led a team with advertising agency ‘Terms One’ from concept to final presentation, architecting an ESG-focused internal software system